Rajiv Gandhi Institute of Petroleum Technology, Jais Amethi

B. Tech 2st year (IT)

Mid-Semester Examination 2022-23

Instructor- Dr. Kalka Dubey

Subject: Object Oriented Methodologies **Course Number**: CS224

Date & Time: 23-09-22, 9:30 AM to 11:30 AM **Total Marks:** 60

INSTRUCTIONS

I. There are 12 questions in this paper.
II. All questions are COMPULSORY.
III. Each question carries 5 Marks only.
IV. Copied answers will not be evaluated.
V. Give a proper explanation for all the questions. Writing only answers directly will not be considered.
VI. Attempt the questions in serial order (if possible).

1. Suppose you give a dinner party for six guests, but your table seats only four. In how many ways can four of the six guests arrange themselves at the table? Any of the six guests can sit in the first chair. Any of the remaining five can sit in the second chair. Any of the remaining four can sit in the third chair, and any of the remaining three can sit in the fourth chair. (The last two will have to stand) So the number of possible arrangements of six guests in four chairs is 6*5*4*3, which is 360. Write a C++ program that calculates the number of possible arrangements for any number of guests and any number of chairs. [5]

2. Differentiate between the pointer and reference in C++ also, give an suitable example of that stated difference between the pointer and references. [5]

3. A library function **islower** takes a single character (a letter) as an argument and returns a non-zero integer if the letter is lowercase, or zero if it is uppercase. This function requires the header file CTYPE.H. Write a C++ program that allows the user to enter a letter, and then displays either zero or non-zero, depending on whether a lowercase or uppercase letter was entered. [5]

4. Define a class **fract** whose object represents rational numbers (**i.e.,** fractions). Inside the **fract** class create two private integers data members named: **numr** and **denr** for storing a numerator and a denominator of a fractional number. Create a parameterized constructor to initialize the data member of **fract** class with the value 0 each. Define the following member function: **eval_fract ()** for evaluating the value of the rational number, **invert ()**

for inverting the rational number, **display ()** for printing the rational number in the form **numr/denr**, **get_numr ()** for returning the values of the **numr** data members, and **get_denr()** returning the values of the **denr** data members, and **reduce_fract ()**, that reduces the fraction numb/denr to the lowest term. Write a C++ program to invoke these member functions in the **main ()** module and test them assuming suitable data from your side. [5]

5. State the use of static data member and static member function of a class. Give a suitable example where we can access the static member function with the name of a class and also create a member function to access all the private data members of this class and initializing them with zero, create one more member function that prints the value of each data member of this class. [5]

6. An electricity board charges the following rates to domestic users to discourage large consumption of energy: [5]

For the first 100 units ₹ 7/- per unit

Beyond 100 units ₹7.5/- per unit

If the total cost of the electricity consumption is more than Rs.250.00 then an additional surcharge of 15% is added to the difference else 10% is added to the difference. Define a class **Electricity** in which all the data members required are private natures and create a member function **count_bill ()** to compute the total cost of electricity consumption units. The total consumption of the electricity unit by the customer is 253. Write a C++ program to calculate the above mention requirement along with the use of appropriate data if needed.

7. Find the true or false of the following questions. Also, give a valid reason (**to be specific not more than one line)**. [1X5]

- **a)** The destructor and constructor are the special member functions that can be called without the dot operator **(.)**.
- **b)** Allocation of memory and frees the previously allocated memory using the new and delete operator respectively in C++.
- **c)** The **sizeof** ()is an operator in C++.
- **d)** C++ provides only private and protected access specifiers for the data members and member functions of a class.
- **e)** Static member function declared inside the class cannot be accessed directly with the class name inside the main function.

8. Explain the use of dynamic initialization of objects and why it is required in object-oriented programming. Illustrate with the help of a suitable example to create some objects and initialize dynamically. [5]

9. Explain the classification of operators in C++, along with the name and symbol of each category operator. [5]

10. Write a C++ program to calculate the following two expressions.

212>>2

212<<4

Mention the result of both the expression in your answer book step by step after the end of the program output. [5]

11. Find out the correct answer to the following multiple-choice question. [1X5=5]

(A) Which among the following is not a necessary condition for constructors?

(i) Its name must be same as that of class

(ii) It must not have any return type

(iii) It must contain a definition body

(iv) It can contain arguments

(B) Which among the following is correct?

(i) class student {public: int student(){} };

(ii) class student {public: void student (){} };

(iii) class student {public: student{}{} };

(iv) class student {public: student(){} };

(C) What is the additional feature in classes that was not in structures?

(i) Data members

(ii) Member functions

(iii) Static data allowed

(iv) Object creation

(D) How many classes can be defined in a single program?

(i) Only 1

(ii) Maximum 1 to 100

(iii) There is no restriction on the number of class creation

(iv) Number of creation of the classes depended mainly on the types of compiler

(E) If data members are private, what can we do to access them from the class object?

(i) Create public member functions to access those data members.

(ii) Create private member functions to access those data members.

(iii) Create protected member functions to access those data members.

(iv) Private data members can never be accessed.

12. Find out the output of the following programs and if there is any compile time error in the program then mention the reason for that error with a valid justification. [2X2.5=5]

(a)

```
#include <iostream>
using namespace std;
class Room
{
int width, length;
public:
void setValue (int w, int h)
{
width = w;
length = h;
}
int getArea ()
 {
   return length * width;
    }
void Display ()
{
cout << "The length of the room is" <<length;
cout << "The width of the room is" <<width;
}
};

int main ()
{
Room objRoom;
objRoom.width=12;
objRoom.length=22;
objRoom.setValue(12,22);
cout << objRoom.getArea () << endl;
objRoom.Display();
return 0;
}
```

(b)

```
#include <iostream>
using namespace std;
class Rectangle
{
  int length, breadth;
   public:
    void setDimension (int l, int b)
    {
      length = l;
       breadth = b;
     }
       int getArea ()
       {
        return length * breadth;
        }
      Rectangle (int x, int y)
      {
       length = x;
       breadth = y;
       cout <<length<<"  " << breadth<<endl;
        }
        ~Rectangle ()
          {
           length = 37;
           breadth = 49;
           cout <<length<<"  " << breadth<<endl;
            }
};

int main ()
{
          Rectangle rt(49,37);
          rt.setDimension (7, 4);
          cout << rt.getArea() << endl;
          return 0;
}
```